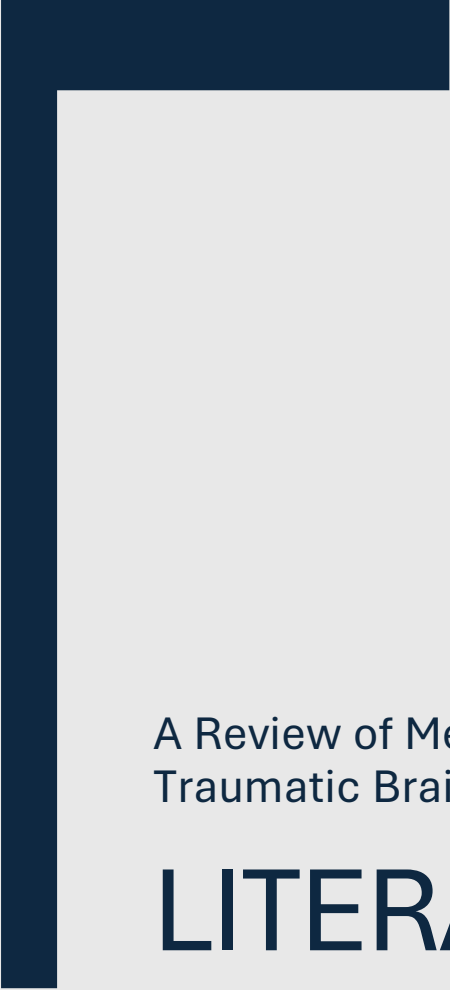




A Review of Methods in Finite Element Analysis and
Traumatic Brain Injury Simulation

LITERATURE REVIEW

Student Intern

Insight Research Services Associated
2025



Table of Contents

1. OVERVIEW	1
1.1. INTRODUCTION.....	1
1.2. METHODOLOGY	1
1.3. FUTURE WORK	2
2. LITERATURE REVIEW	4
2.1. MATHEMATICAL MODELING OF THE HUMAN BRAIN: FROM MAGNETIC RESONANCE IMAGES TO FINITE ELEMENT SIMULATION	4
2.2. AN ANALYTICAL REVIEW OF THE NUMERICAL METHODS USED FOR FINITE ELEMENT MODELING OF TRAUMATIC BRAIN INJURY	9

1. Overview

1.1. Introduction

This document reviews the role of numerical and data-driven methods in computational brain modeling, with a focus on linear algebra techniques, finite element modeling (FEM), and surrogate modeling approaches for simulating traumatic brain injury. The analysis builds on two key works: *Mathematical Modeling of the Human Brain: From Magnetic Resonance Images to Finite Element Simulation* and *An Analytical Review of the Numerical Methods used for Finite Element Modeling of Traumatic Brain Injury*.

The literature review is organised into two main sections. The first examines how linear algebra supports brain modeling, particularly in mesh preprocessing and adjustment for MRI-based simulations. The second explores FEM techniques in head injury research, covering mesh generation, material modeling, and computational efficiency improvements. By analysing citations of important papers, this review synthesises recent advancements that refine or extend these methodologies.

These findings will provide a foundation for future research, offering interns a starting point for determining which numerical and data-driven techniques are most efficient and practical for brain modeling.

1.2. Methodology

Section 2 of this literature review is divided into two parts, Section 2.1 and Section 2.2, each presenting a comprehensive literature review table summarising the selected studies from the two focal books.

Section 2.1 examines the applications of linear algebra in computational brain modeling as presented in *Mathematical Modeling of the Human Brain: From Magnetic Resonance Images to Finite Element Simulation*. A detailed reading of the text revealed several implementations, with a particular focus on the “Improved Volume Meshing by Surface Preprocessing” section (Section 3.2, Page 28). In this section of the book, the process of mesh preprocessing is described, where techniques such as remeshing, smoothing (using methods like Taubin or Laplacian smoothing), and defect repair are employed to enhance simulation accuracy and stability. Linear algebraic operations underpin these processes through vertex position adjustments via matrix operations and the representation of mesh connectivity using adjacency matrices. Following the identification of this key application, the reference that underpins the mesh preprocessing approach was traced to the FreeSurfer topological defects tutorial guide, specifically the FreeSurfer Wiki (2020) available at <https://surfer.nmr.mgh.harvard.edu/fswiki>. To explore how this foundational

methodology has been advanced in more recent research, a citation analysis was conducted using Scopus. The FreeSurfer Wiki URL was entered into the "Search documents" box with the "References" option selected, ensuring that the search focused on documents citing this URL, and a publication date filter from 2015 onward was applied to capture recent advancements over the last decade. The titles and abstracts of the resulting 111 documents were then carefully reviewed to filter out studies that employed FreeSurfer in a conventional manner, retaining only those that introduced new or alternative methods. This process resulted in the selection of 15 papers, which are detailed in the table in Section 2.1.

Section 2.2 focuses on identifying and analysing papers related to *An Analytical Review of the Numerical Methods used for Finite Element Modeling of Traumatic Brain Injury*. The review process was divided into two parts. The first part involved an exhaustive screening of the 96 references cited by the book using Scopus. Each reference's title and abstract was examined for its relevance to FEM methodology, with a specific emphasis on contributions that advanced the mathematical underpinnings of FEM. The selection criteria required that a reference distinctly push the boundaries of FEM or numerical techniques rather than merely applying an existing FE model, which resulted in the identification of 19 papers. In the second part, a citation search was conducted using Scopus to retrieve documents that cited the book, yielding 67 results. These documents were then reviewed based on their titles and abstracts to determine whether they proposed or tested new computational or mathematical methods, or introduced software innovations for FE head modeling or head-impact modeling. Only studies that significantly advanced the underlying methodology, through improvements in numerical efficiency, reduced simulation run-times, or novel computational frameworks, were retained, leading to the selection of 15 papers, which are compiled in the table in Section 2.2.

All selected studies from both books have been compiled into two separate tables, one for each section, that include bibliographic details, methodological commentary, and specific software or implementation details.

1.3. Future Work

The next phase of this project will involve a practical evaluation of the computational efficiency of the various numerical techniques identified in the literature. The objective is to determine which methods can be implemented on a standard computer to compare their run-times and overall performance. Based on the literature review, several software packages and libraries are readily accessible and can be installed locally. For instance, FreeSurfer is freely available for download and is widely used for brain segmentation and mesh generation. Complementary Python libraries such as scikit-learn and UMAP

(umap-learn) provide accessible tools for dimensionality reduction and clustering, while PyTorch offers a robust framework for implementing deep learning models that can serve as surrogates for computationally intensive FE simulations.

In addition, open-source tools like ANTs for image registration, FSL for preprocessing, and Python-based alternatives (e.g., DIPY for diffusion imaging analyses) are viable options for integration into the computational pipeline. Conversely, several methods identified in the literature rely on high-performance computing (HPC) resources or proprietary frameworks, making them challenging to replicate on a standard computer. For example, approaches utilising big data platforms like Hadoop/Spark or advanced clusters require specialised hardware and complex setup procedures. These methods, while offering valuable insights and benchmarks, will be acknowledged as reference points for future research but will not be the primary focus of the initial comparative study. Similarly, some specialised commercial FE solvers such as LS-DYNA and Abaqus require costly proprietary licenses and are less accessible on personal computers; therefore, the initial focus will be on those methods that leverage open-source platforms and readily available software.

Overall, future work will prioritise implementing and benchmarking techniques that can be executed on accessible hardware. This approach will facilitate direct comparisons of computational times and model accuracy across different methods, such as mesh preprocessing, deep learning-based surrogate modeling, and alternative FE methodologies, and serve as a reproducible blueprint for future students interested in advancing computational modeling in traumatic brain injury research. The eventual goal is to set up these environments, benchmark key processes, and document the workflow to provide guidance for subsequent research and implementation efforts.

Furthermore, the papers highlighted in Sections 2.1 and 2.2 will serve as "seed articles" for future literature review efforts. These selected papers represent significant methodological advances in both linear algebra applications for brain modeling and FEM techniques for traumatic brain injury simulation. Future students can use citation tracking tools like Scopus, Web of Science, or Google Scholar to identify new papers that have cited these seed articles, enabling them to discover the latest methodological improvements and applications. This approach to literature monitoring will help maintain an up-to-date understanding of the field's evolution, particularly in areas such as computational efficiency improvements, novel numerical methods, and innovative software implementations. Regular citation analysis of these seed articles will also help identify emerging research trends and potential new directions for computational brain modeling. It is recommended to conduct such citation analyses at least every six months to capture new developments in this rapidly evolving field. This approach will ensure future research builds upon the most current methodological advances.

2. Literature Review

2.1. Mathematical Modeling of the Human Brain: From Magnetic Resonance Images to Finite Element Simulation

Document Title	Authors	Year	Cited By	Comments	Software
Cited FreeSurfer from <i>Mathematical Modeling of the Human Brain: From Magnetic Resonance Images to Finite Element Simulation</i>					
<i>Predicting cognitive decline in a low-dimensional representation of brain morphology</i>	Lamontagne-Caron et al.	2023	0	Utilises a novel nonlinear dimension reduction technique (UMAP) on cortical thickness data obtained from FreeSurfer. Embeds data into a 2D manifold to compare groups and trajectories, surpassing standard morphometric approaches. Facilitates the prediction of cognitive decline by analysing brain morphology trajectories.	▪ FreeSurfer version 6.0: Used to quantify brain anatomy and extract cortical thickness measures (executed on the CBRAIN platform). ▪ scikit-learn: Used for K-means clustering. ▪ UMAP (Python umap-learn library): Possibly used for dimension reduction (not explicitly stated in the paper).
<i>Joint-label fusion brain atlases for dementia research in Down syndrome</i>	Queder et al.	2022	0	Develops disease-specific brain atlases for Down syndrome using joint-label fusion. Compares newly built DS atlases to a standard atlas for PET region quantification, providing a novel approach for specialised atlas creation.	▪ Advanced Normalization Tools (ANTs) version 2.2.0: Used for joint label fusion atlas creation. ▪ FreeSurfer version 6.0: Used to generate individual MRI-based segmentations.
<i>Prediction of Alzheimer's progression based on multimodal Deep-Learning-based fusion and visual Explainability of time-series data</i>	Rahim et al.	2023	48	Combines multimodal data (MRI + clinical scores) using a 3D Convolutional Neural Network (CNN) followed by a Bidirectional Recurrent Neural Network (BRNN). Introduces a visual Explainability (XAI) mechanism to highlight significant brain regions, advancing beyond simple region-based volumetry.	▪ FSL (version not specified): Used for visualisation, reorientation, skull stripping, and MNI152 registration (FSLEyes, fslreorient2std, FLIRT, BET2). ▪ FreeSurfer (version not specified): Used for brain surface extraction and manual segmentation. ▪ PyTorch version 1.7.0: Used to implement the main deep learning model. ▪ MATLAB (BrainNet Viewer): Used for brain surface rendering. ▪ Python (MedCam library): Used for visualising salient features contributing to the output class determination.
<i>Fully Automatic Adaptive Meshing Based</i>	van Doormaal et al.	2021	11	Describes a cloud-based fully automatic adaptive meshing algorithm for T1 MRI segmentation of ventricles and direct augmented reality (AR)	▪ 3D Slicer version 4.10.2: Used for image analysis and manual segmentation to create the ground truth dataset.

<i>Segmentation of the Ventricular System for Augmented Reality Visualization and Navigation</i>				visualisation for neurosurgery. This specialised pipeline replaces standard mesh extraction, enabling real-time navigation.	▪ MATLAB (custom script): Used to perform calculations for the adaptive meshing algorithm. ▪ SPSS (version not specified): Used for statistical analyses
<i>Extensive T1-weighted MRI preprocessing improves generalizability of deep brain age prediction models</i>	Dular et al.	2024	2	Examines how different T1 preprocessing pipelines (registration transforms, grayscale correction, etc.) impact deep-learning brain-age models. Concludes that heavier preprocessing often yields better generalisability, contradicting some assumptions about minimal preprocessing.	▪ PyTorch version 1.4.0 (Python version 3.6.8): Used to implement deep-learning brain-age models. ▪ NiftyReg (version not specified): Used for registration and resampling in the RIG, RIG+GS, and AFF+GS pipelines. ▪ FreeSurfer (version not specified): Used for grayscale correction. ▪ FSL FLIRT (version not specified): Used for registration in the Fs+FSL pipeline. ▪ N4 algorithm: Used for inhomogeneity correction in some pipelines.
<i>An Optimized Deep Learning Model for Predicting Mild Cognitive Impairment Using Structural MRI</i>	Alyoubi et al.	2023	4	Employs deep CNN feature extraction (VGG16, Inception-V3, ResNet50) specifically on entorhinal cortex MRI slices for MCI detection. Focuses on a small ROI plus advanced optimisation of deep learning.	▪ SPM12: Used for aligning MRI images to a standard template. ▪ FreeSurfer (version not specified): Used for 3D surface construction and extraction of the entorhinal cortex with the Desikan–Killiany atlas. ▪ Pre-trained CNN models (VGG16, Inception-V3, ResNet50): Used for feature extraction from the EC area. ▪ CNN with Sigmoid Activation & SVM (scikit-learn): Used for classification. ▪ Pillow (Python library): Used for MRI slice analysis.
<i>Deep learning volumetric brain segmentation based on spectral CT</i>	Fransson et al.	2023	1	Investigates a deep learning approach to volumetric segmentation of gray/white matter from spectral CT data, not conventional MRI. Novel pipeline exploring CT-based brain mapping, which is unusual in typical FreeSurfer contexts.	▪ Not explicitly specified: Likely Python-based deep learning framework or specialised research tools (cannot access full text).
<i>Characterizing Streamline Count Invariant Graph Measures of Structural Connectomes</i>	Newlin et al.	2023	3	Focuses on graph-theoretical connectome metrics derived from diffusion MRI data and their invariance to streamline count in tractography. Uses FreeSurfer for parcellation but introduces new network analysis to handle connectivity data reliability and reproducibility.	▪ MRtrix version 3.0.0: Used for tractography and connectome generation. ▪ FSL TOPUP/eddy version 3.0.3: Used for eddy current correction, motion correction, and EPI/gradient nonlinearity distortion correction.

					▪ Brain Connectivity Toolbox (BCT) version 03/03/2019 (MATLAB-based): Used for most graph measures. ▪ BCTpy (Python-based extension): Used for randomness, small-worldness, and edge count graph measures. ▪ MATLAB version 3.0.3: Used for statistical analyses.
<i>Three-dimensional histology reveals dissociable human hippocampal long-axis gradients of Alzheimer's pathology</i>	Ortega-Cruz et al.	2024	1	Presents "Path2MR" - a pipeline for 3D histology reconstruction using blockface photos (and minimal MRI) aligned to an atlas. This merges postmortem histology with MRI-based standard space, offering a novel approach beyond typical structural MRI segmentation.	▪ FreeSurfer version 7.4: Used for photograph processing and 3D reconstruction. ▪ GIMP version 2.10.28: Used for slice segmentation. ▪ NiftyReg (version not specified): Used for nonlinear registration to the MNI atlas. ▪ Kernel Regression: Used for histology-to-atlas registration (implementation details not specified). ▪ FSL tools (fslslice, fslmaths, fslstats): Used for analysing intensity measures. ▪ CellProfiler version 4.0.7: Used for quantifying tau and Aβ burden. ▪ RStudio version 1.4.1106: Used with packages (ggpubr, tidyverse, readxl, rstatix, viridis) for statistical analysis. ▪ https://github.com/ortegacruz/Path2MR
<i>Degree of improving TMS focality through a geometrically stable solution of an inverse TMS problem</i>	Makarov et al.	2021	8	Proposes a finite/boundary element approach for the inverse TMS problem to maximise focality. Incorporates advanced HPC-based geometric solvers and numerical methods, significantly advancing beyond standard cortical thickness measures.	▪ SimNIBS 3.1: Used for brain segmentation and meshing (includes mri2mesh and headreco). ▪ FreeSurfer & FSL: Used within SimNIBS for anatomical segmentation. ▪ SPM/CAT: Used within SimNIBS's headreco module for structural brain mapping. ▪ BEM-FMM Forward Solver (MATLAB implementation): Used for solving the forward TMS problem, implemented in MATLAB.
<i>Cloud Based Big Data Platform for Image Analytics</i>	Vuppala et al.	2017	8	Describes a cloud-based big-data pipeline for medical image analytics across multiple institutions, including advanced de-identification and a clinical decision support system. Combines big data, HPC, and advanced analytics.	▪ Hadoop/HDFS: Used for distributed storage of large-scale data. ▪ MapR-DB (NoSQL): Used for managing DICOM files efficiently. ▪ Mesosphere: Used for orchestrating and managing computational resources.

					▪ Apache Spark: Used for large-scale data processing and analytics. ▪ Java, Scala, Python: Used for anonymisation and image-processing tasks. ▪ Finagle, SBT, JUnit, ScalaTest: Used for building and testing the platform. ▪ Apache Commons: Currently being integrated to enhance functionality.
<i>Medical IT service for the detection of neurocognitive pathologies based on the results of an MRI scan of the human brain</i>	Enyagina et al.	2022	0	Develops a web-based service for automated detection of multiple neurocognitive pathologies (autism, schizophrenia, stroke risk). Emphasises the creation of a “3D mathematical model” from MRI data and an integrated online front-end, offering a comprehensive medical IT pipeline beyond standard volumetry with FreeSurfer.	▪ Front-End (JavaScript, HTML): Used for the interactive web user interface. ▪ BrainBrowser: Used for primary online visualisation of MRI data. ▪ Back-End (Java, J2EE): Used for server-side development. ▪ Spring, Hibernate: Used for building scalable enterprise applications. ▪ Apache, Tomcat: Used for managing HTTP requests and hosting the web application. ▪ BPMN 2.0, Activiti: Used for workflow modeling and management. ▪ MySQL version 5.5: Used for storing experiment metadata and user data. ▪ MRIConvert: Used to convert DICOM files into suitable processing formats. ▪ FreeSurfer, FreeView (versions not specified): Used for preprocessing, segmentation, and 3D modeling of MRI data. ▪ SBT, JUnit, ScalaTest: Used for building and testing software components. ▪ XML: Used for data representation and configuration.
<i>VRain: Virtual reality assisted intervention for neuroimaging</i>	Duncan et al.	2017	6	Presents a virtual reality (VR) editing environment for MRI segmentations derived from FreeSurfer. Instead of conventional 2D slice-by-slice corrections, it leverages VR to allow users to edit multiple slices simultaneously in a 3D space, significantly enhancing standard correction workflows.	▪ Unity: Used as the game engine for VR interface development. ▪ NIFTI-1: Used as the MRI data format for processing and visualisation. ▪ Look Up Tables (LUTs): Used for mapping MRI scalar values to colours within the VR environment. ▪ Cubiquity Voxel Engine: Initially explored for rendering MRI data (later replaced).

					▪ Multi-layered Cube Approach: Adopted for improved rendering and user interaction. ▪ VR Wand: Used as an input device for precise MRI segmentation edits in VR.
<i>Automatic group-wise whole-brain short association fiber bundle labeling based on clustering and cortical surface information</i>	Vázquez et al.	2020	3	Implements an automated fibre clustering method for short-association bundles, guided by cortical connectivity. This approach merges cluster analysis of tractography with a robust surface atlas (Desikan–Killiany). It advances typical diffusion pipelines by focusing on short, superficial white matter (WM) bundles, enhancing the precision and reliability of fibre bundle labeling.	▪ Python version 3.6: Used for pipeline development and data processing. ▪ QuickBundles (DIPY library): Used for fiber clustering based on streamline similarity. ▪ Hungarian Algorithm (scipy): Used for optimising the matching of clusters across subjects. ▪ BrainVISA/Connectomist version 2.0: Used for artifact and distortion correction in diffusion MRI data. ▪ FreeSurfer (version not specified): Used for generating cortical meshes and anatomical labeling (Desikan–Killiany atlas).
<i>System for storing and analyzing experimental MRI/fMRI data on the hybrilit heterogeneous platform</i>	Zuev et al.	2020	0	Describes a cloud/HPC-based system (“HybriLIT”) for large-scale MRI/fMRI data storage and analysis. Adopts a “digital laboratory” approach that integrates specialised software and automates parts of the imaging workflow, extending beyond standard single-lab usage to support multi-institutional collaboration and large-scale data processing.	▪ MRIConvert: Used to convert DICOM files to NIFTI format. ▪ FreeSurfer: Used for cortical mesh generation and anatomical labeling of MRI data. ▪ FSL: Used for NIFTI file processing such as artifact correction. ▪ FreeView (from FreeSurfer): Used for visualising processed MRI/fMRI data. ▪ Modules Package: Used for managing environment variables on HPC resources. ▪ SLURM Scheduler: Used for job management on the HybriLIT platform.

2.2. An Analytical Review of the Numerical Methods used for Finite Element Modeling of Traumatic Brain Injury

Document Title	Authors	Year	Cited By	Comments	Software
References Used in An Analytical Review of the Numerical Methods used for Finite Element Modeling of Traumatic Brain Injury					
<i>Mesh considerations for finite element blast modelling in biomechanics</i>	Panzer et al.	2013	19	Examines mesh design (tetra vs. hexa) and element size effects on blast simulations, performs mesh-ratio and sensitivity analyses, and highlights fluid-structure interaction considerations in shock-tube problems.	▪ LS-DYNA version 971.R5: Used for fluid-structure interaction modeling in a shock tube. ▪ HyperMesh version 10.0: Used to generate hexahedral and tetrahedral meshes.
<i>Automatic generation and validation of patient-specific finite element head models suitable for crashworthiness analysis</i>	Ho et al.	2009	20	Presents a fully automated segmentation-to-meshing pipeline for creating subject-specific head models, validates against cadaveric test data, and discusses element quality and head-geometry fidelity for crashworthiness simulations.	▪ Custom Java algorithm: Used for FE model generation from T1-weighted MRI images. ▪ Expectation Maximisation Segmentation (EMS): Used for tissue classification. ▪ LS-DYNA: Used for simulating and validating the FE models against PMHS head impacts.
<i>Prediction of brain deformations and risk of traumatic brain injury due to closed-head impact: quantitative analysis of the effects of boundary conditions and brain tissue constitutive model</i>	Wang et al.	2018	45	Investigates how different skull-brain interface models (including CSF and meninges) and constitutive laws (Ogden vs. linear) affect intracranial strain, concluding that realistic interface modeling and hyperviscoelasticity improve TBI predictions.	▪ LS-DYNA 971: Nonlinear explicit FE code utilised to simulate head-neck impacts on post-mortem human subjects (PMHSs) using the THUMS Version 4.0 model. ▪ THUMS Version 4.0: Employed as the human body model for FE simulations in LS-DYNA.
<i>Development and validation of a new finite element human head model: Yet another head model (YEAHM)</i>	Fernandes et al.	2018	50	Creates a detailed anatomical head mesh accounting for sulci/gyri, utilises automatic meshing from imaging, assigns material properties, and validates predictions against intracranial pressure data from frontal and occipital impacts.	▪ OsiriX (MIA plugin): Used for segmenting CT and MRI images. ▪ Meshmixer: Used for smoothing the triangular meshes. ▪ Meshlab: Used for repairing and closing gaps in the triangular meshes. ▪ CATIA V5: Used for generating solid CAD models from the segmented meshes. ▪ Abaqus version 6.13: Used for FE meshing and explicit dynamic simulations.

<i>Parametric comparisons of intracranial mechanical responses from three validated finite element models of the human head</i>	Ji et al.	2014	91	Compares three validated head models (DSNM, SIMon, WSUHIM) under identical impacts, highlights how geometry, discretisation, and material parameters influence intracranial strain/stress, and cautions about mixing results from different FE models.	▪ Abaqus version 6.12: Used for DSNM simulations. ▪ LS-DYNA: Used for SIMon and WSUHIM simulations. ▪ TrueGrid: Used for meshing in the SIMon model. ▪ HyperMesh: Used for meshing in the WSUHIM model.
<i>Development of a finite element human head model partially validated with thirty five experimental cases</i>	Mao et al.	2013	322	Involves extensive validation across 35 experimental impacts (frontal, lateral, oblique), incorporates multi-layer skull, bridging veins, and advanced brain material definitions, and discusses hourglass control, damping, and numerical stability for real-world simulations.	▪ HyperMesh version 10.0: Used for skull meshing based on CAD data from MRI/CT. ▪ ANSYS ICEM CFD/HEXA version 12.0: Used for brain meshing (hexahedral). ▪ LS-DYNA MPP: Used for simulations and validation of the FE head model.
<i>Investigation of traumatic brain injuries using the next generation of simulated injury monitor (SIMon) finite element head model</i>	Takhounts et al.	2008	325	Upgrades the SIMon model with refined geometry and constitutive laws, derives CSDM and other strain-based injury metrics, and validates with cadaver data and mild TBI impacts, comparing rotational vs. translational kinematics.	▪ TrueGrid: Used for meshing the skull, dura-CSF, and brain from CAD data. ▪ LS-DYNA: Used for conducting simulations and comparing with experimental data.
<i>Evaluation of head injury criteria using a finite element model validated against experiments on localized brain motion, intracerebral acceleration, and intracranial pressure</i>	Kleiven	2006	219	Presents the KTH head model validated using intracranial pressure and brain displacement data, evaluates how well HIC, HIP, etc. correlate with FE-based strain, and highlights the influence of rotational kinematics for severe TBI.	▪ LS-DYNA: Used to simulate a detailed FE head model validated against experimental data.
<i>Development and validation of an advanced anisotropic visco-hyperelastic human brain FE model</i>	Sahoo et al.	2014	109	Incorporates diffusion tensor imaging (DTI) for white-matter anisotropy, uses visco-hyperelastic tissue laws, validates with head impact experiments (pressure and brain motion), and shows improved predictions of local strain in fibre tracts.	▪ Radioss: Initially used for the Strasbourg University Finite Element Head Model (SUFEHM). ▪ LS-DYNA: Used to conduct impact simulations using *MAT_092_SOFT_TISSUE_VISCO. ▪ MATLAB version 7.4: Used to derive anisotropic material properties from DTI data.
<i>Development of a metric for predicting brain strain responses using head kinematics</i>	Gabler et al.	2018	92	Proposes the universal brain injury criterion (UBrIC) correlating rotational velocity with FE-derived strain, uses 1,600 simulated impacts of varied severity/direction, and shows that rotational velocity is a strong predictor of max principal strain.	▪ LS-DYNA version 971 R7.1.1 (double precision): Used for FE head model simulations (GHBMCM50 and SIMon v4.0).

					▪ MATLAB version 8.4.0: Used for statistical analysis (fitting kinematic metrics to strain metrics).
<i>A brain impact stress analysis using advanced discretization meshless techniques</i>	Marques et al.	2018	7	Proposes a meshless discretisation (Radial Point Interpolation Method, RPIM) for simulating head/brain impacts, contrasting it with classical FEM. Demonstrates how meshless approaches can provide higher convergence rates and smoother stress/strain fields. Performs direct comparisons to standard FE analyses to show advantages in stability and accuracy.	▪ In-house or research code implementing RPIM: Used to perform meshless head-impact simulations (cannot access full text).
<i>Uncertainty quantification for constitutive model calibration of brain tissue</i>	Brewick et al.	2018	22	Utilises Bayesian estimation (Hamiltonian MCMC) to calibrate Ogden-type constitutive models under Drucker stability constraints. Emphasises how confidence bounds on parameters propagate to predicted tissue stresses and strains. Showcases a systematic uncertainty quantification framework for soft-tissue FE models.	▪ Hamiltonian Monte Carlo Implementation: Likely an in-house or standard library code in MATLAB, R, or Python (not specified in the paper).
<i>White Matter Anisotropy for Impact Simulation and Response Sampling in Traumatic Brain Injury</i>	Zhao et al.	2019	59	Compares different anisotropic implementations (voxel- vs. tractography-based) for white-matter modeling. Investigates how varying fibre-reinforcement strategies affect local strain metrics, using a parametric approach. Re-validates head models for multiple loading scenarios, demonstrating how anisotropic detail can alter brain-injury risk predictions.	▪ Abaqus/Explicit version 2016: Used for FE simulations of the modified Worcester Head Injury Model (WHIM). ▪ MATLAB version R2017b: Used for pre-processing and post-processing tasks (data analysis, visualisation).
<i>Modeling the mechanics of axonal fiber tracts using the embedded finite element method</i>	Garimella et al.	2017	55	Presents an embedded FE approach, representing axonal tracts with 1D fibre elements placed into the standard 3D brain mesh. Avoids re-meshing the solid domain. Demonstrates improved axonal strain calculations vs. isotropic FE. Verification includes relative displacement data and parametric studies.	▪ Diffusion Toolkit & TrackVis: Used for constructing and visualising tractography data. ▪ dcm2nii: Used for converting DICOM images to NIfTI format. ▪ ANSYS ICEM CFD version 14.5.7: Used for FE mesh generation. ▪ MATLAB (custom script): Used to convert tractography data into FE truss elements and for data analysis. ▪ Abaqus Explicit version 6.13-2: Used for embedded FE method simulations.) to model axonal fibre tracts.

<i>A highly efficient enhanced assumed strain physically stabilized hexahedral element</i>	Puso	2000	112	Describes a single-point integrated hex element with enhanced assumed strains (EAS) and “physical stabilisation” to prevent hourglass instabilities. Retains accuracy and efficiency for large-strain dynamic simulations common in crash or impact scenarios.	▪ In-house or research FEM code: The EAS formulation was likely tested in a custom solver or early version of LS-DYNA (not explicitly stated).
<i>Comparison of hexahedral and tetrahedral elements in finite element analysis of the foot and footwear</i>	Tadepalli et al.	2011	146	Although foot-specific, it provides a systematic mesh-convergence and error analysis of hex vs. tet meshes (linear and quadratic). Demonstrates accuracy vs. computational cost tradeoffs, relevant to modeling soft tissues (like the brain). Contributes quantitative guidelines on element types for biomedical FE simulations, including nearly incompressible regions.	▪ Abaqus Standard version 6.10: Used for FE analyses, including meshing and solving simulations.
<i>A simple average nodal pressure tetrahedral element for incompressible and nearly incompressible dynamic explicit applications</i>	Bonet et al.	1998	221	Introduces a tetrahedral element specifically designed to handle incompressible/near-incompressible materials by averaging nodal pressures, thus reducing volumetric locking. Showcases good performance in large-strain explicit dynamics problems.	▪ In-house FEM code (not explicitly stated): The method was likely implemented in a custom solver (not explicitly stated).
<i>A stabilization procedure for the quadrilateral plate element with one-point quadrature</i>	Belytschko et al.	1983	232	A foundational reference on hourglass stabilisation for under-integrated elements. Describes how to systematically remove spurious zero-energy modes, ensuring correct strain fields without excessive computational cost. Forms the basis of many explicit FE codes’ quadrilateral/hexahedral element formulations used in crash and brain-injury simulations.	▪ In-house code (Northwestern University, early 1980s): Influenced modern codes like LS-DYNA, ABAQUS/Explicit, etc. (cannot access full text).
<i>Smoothed Finite Element Methods (S-FEM): An Overview and Recent Developments</i>	Zeng et al.	2018	261	Surveys various S-FEM formulations (e.g., CS-FEM, ES-FEM, NS-FEM) that modify strain-field integration to improve accuracy and convergence in linear or nonlinear problems. Useful for nearly incompressible or high-deformation analyses. Demonstrates how S-FEM can reduce stiffness and locking issues, especially relevant in soft-tissue FE modeling.	▪ Custom codes (MATLAB or C++) / User-defined subroutines (ABAQUS): Typical implementations of S-FEM.
Cited An Analytical Review of the Numerical Methods used for Finite Element Modeling of Traumatic Brain Injury					
<i>The Impact of Drop Test Conditions on Brain Strain Location and Severity: A Novel</i>	Stilwell et al.	2024	0	Proposes a CNN-based approach to predict brain strains under drop-test scenarios, replacing many repetitive FE simulations; the trained network provides near-instant estimates of strain for new impact settings.	▪ Pre-trained CNN (from GitHub): Used to estimate regional brain strain. https://github.com/Jilab-biomechanics/CNN-brain-strains

<i>Approach Using a Deep Learning Model</i>					▪ LabVIEW: Used for recording data from the nine-accelerometer package during drop tests. ▪ MATLAB: Used for post-processing recorded kinematic data. ▪ Python version 3.8 (scipy.signal): Used for additional data filtering. ▪ R: Used for statistical analysis (two-way ANOVA, Tukey's HSD).
<i>Physics-Informed UNets for Discovering Hidden Elasticity in Heterogeneous Materials</i>	Kamali et al.	2024	6	Introduces a UNet-based physics-informed neural network for inversion in elasticity problems; offers faster and more accurate property reconstruction than fully connected physics-informed neural networks.	▪ ANSYS Workbench: Used for FE simulations generating training data. ▪ PyTorch version 1.13.1: Used for developing the inverse models (EI-UNet). ▪ MATLAB: Used for pre-processing FE simulation data (e.g., interpolation from unstructured to structured grids). ▪ https://github.com/klaksari/2D_EI-UNet
<i>Predicting Shock-Induced Cavitation Using Machine Learning: Implications for Blast-Injury Models</i>	Marsh et al.	2024	0	Uses machine learning (k-nearest neighbour, SVM) to predict shock-induced cavitation bubble counts, reducing reliance on expensive fluid–structure simulations and demonstrating a fast surrogate modeling approach.	▪ MATLAB version 9.10.0.1602886 (R2023a): - Custom Bubble Detection Program: Automatically identifies and tracks cavitation bubbles. - fitcknn Function: Implements/cross-validates k-NN models. - fitcecoc Function: Implements/cross-validates ECOC SVM models. - cvpartition / crossval: Performs cross-validation.
<i>A Finite Rotation, Small Strain 2D Elastic Head Model, With Applications in Mild Traumatic Brain Injury</i>	Wan et al.	2023	4	Presents a 2D head model that accounts for finite skull rotation and small strains in the brain, enabling simpler and faster computations for mild TBI risk analysis than full 3D FE methods.	▪ Custom numerical integration scripts: Likely implemented in MATLAB or Python (not explicitly stated).
<i>An Image Registration–Based Morphing Technique for Generating Subject-Specific Brain Finite Element Models</i>	Giudice et al.	2020	30	Describes an automated mesh-morphing pipeline via deformable image registration to quickly produce subject-specific FE brain models, drastically reducing manual mesh setup.	▪ Advanced Normalization Tools (ANTs): Used for image registration and brain tissue segmentation. ▪ MATLAB (custom script): Used for generating voxel-based FE models from segmented images.

					▪ LS-DYNA version mpp971R9.1.0 (double precision): Used for FE simulations based on voxel models.
<i>An Anatomically Detailed and Personalizable Head Injury Model: Significance of brain and white matter tract morphological variability on strain</i>	Li et al.	2021	82	Leverages hierarchical multiple-feature registration to morph a baseline hex-based head model to new MRIs, capturing large anatomical differences efficiently and thereby cutting re-meshing time.	▪ 3D Slicer version 3.6: Used for T1W/T2W MRI segmentation and Demons registration. ▪ Hexotic: Used for generating hexahedral meshes with an octree algorithm. ▪ LS-DYNA version 971 R11 (explicit dynamic solving method): Used for FE simulations of head impacts. ▪ FSL version 6.0.2 (DTIFit): Used for diffusion tensor imaging processing. ▪ FreeSurfer version 7.1.0: Used for automated segmentation (Dice coefficient calculation). ▪ Dramms version 1.5.1: Used for deformable image registration. ▪ Custom Python code: Adapted for specific data processing tasks. ▪ CORA version 3.6.1: Used for correlation analysis (simulation metrics vs. reference data).
<i>Rapid Prediction of Brain Injury Pattern in mTBI by Combining FE Analysis With a Machine-Learning Based Approach</i>	Shim et al.	2020	18	Combines a library of precomputed FE data with partial least squares regression to efficiently predict local brain strain patterns in near real time, significantly reducing the computational cost of repeated FE simulations.	▪ OpenCMISS: Used for creating subject-specific FE brain models (including white matter tracts). ▪ Python (SciPy, scikit-learn): Used for training Partial Least Squares Regression (PLSR) models to predict strain based on impact parameters.
<i>Prediction of skull fractures in blunt force head traumas using finite element head models</i>	Lindgren et al.	2024	1	Introduces a strain-rate-dependent cranial bone material model and applies it to subject-specific FE head models built through image-based morphing. Validates against post-mortem CT and autopsy data to reproduce real-world fracture patterns. Demonstrates refined capabilities in predicting skull fractures for blunt impacts using advanced constitutive formulations and subject-personalised head models, contributing to forensic and clinical investigations.	▪ LS-PrePost version 4.8: Used for preparing simulation inputs and FE models. ▪ LS-DYNA version 13: Used for conducting FE simulations of head impacts. ▪ MATLAB version 2021a: Used for analysing simulation outputs and validation. ▪ 3D Slicer version 4.11: Used for segmenting PMCT scans.

					▪ Hexotic: Used for meshing skullcap specimens with solid hexahedral elements. ▪ Demons registration (3D Slicer BRAINSdemonWarp): Used for non-linear image registration to create subject-specific head models.
<i>Full-field, frequency-domain comparison of simulated and measured human brain deformation</i>	Arani et al.	2024	0	Presents a frequency-domain framework for comparing FE brain deformation simulations to full-field magnetic resonance elastography (MRE) data collected in vivo under harmonic skull motion. Enables robust voxel-wise matching of strain fields while accounting for magnitude and phase, thereby improving calibration and evaluation of computational head models with respect to high-resolution MRE measurements.	▪ ABAQUS version 2021: Used for FE simulations under harmonic skull motion. ▪ MATLAB version R2023a: Used for quantitative comparisons to MRE data. ▪ FSL PRELUDE: Used for phase data unwrapping in MRE experiments. ▪ ANTs: Used for nonlinear registration of strain fields to MNI-152 space.
<i>Rheology of brain tissue and hydrogels: A novel hyperelastic and viscoelastic model for forensic applications</i>	Yarin et al.	2023	6	Proposes a rational-function-based strain-energy approach for hyperelastic modeling of soft tissues, including strong compressibility and nonlinear response, and expands this with a Bernstein-Kearsley-Zapas viscoelastic framework suitable for high-rate forensic scenarios. Offers a new material model spanning large deformation and fading-memory effects, facilitating realistic simulations of brain and hydrogel-like tissues under ballistic or blast conditions, targeting forensic injury analyses.	▪ Wolfram Mathematica: Used for analytical derivations of the constitutive model. ▪ In-house code (C++/Fortran): Likely used for implementing the hyperelastic and viscoelastic models (not explicitly stated).
<i>Concussion Prone Scenarios: A Multi-Dimensional Exploration in Impact Directions, Brain Morphology, and Network Architectures Using Computational Models</i>	Wu et al.	2022	5	Integrates a conventional FE head model with neural network dynamics to investigate how mechanical deformation can alter functional connectivity during potentially concussive impacts. Explores how variations in impact direction, anatomical size, and axonal network architecture influence tissue strain and network-level responses, illustrating a more mechanobiological view of concussion that combines structural and functional modeling for better injury insights.	▪ LS-DYNA version 971 R9.2.0: Used for FE simulations of brain deformation. ▪ MATLAB version 2019b: Used for neurodynamic/hemodynamic simulations (Runge-Kutta) and data analysis.
<i>Subject-specific finite element head models for skull fracture evaluation—a new tool in forensic pathology</i>	Henningsen et al.	2024	1	Develops subject-specific FE head models from post-mortem computed tomography (PMCT) and image morphing to reconstruct blunt-force events. Demonstrates the feasibility of simulating skull	▪ LS-DYNA version 13: Used for FE simulations of head impacts. ▪ LS-PrePost version 4.8: Used for pre-/post-processing FE models.

				fractures in routine forensic pathology by testing how well the numerical results replicate real-world autopsy fractures. Represents an advancement in subject-specific pipelines but relies on standard FE approaches for the mechanical solver.	▪ 3D Slicer version 4.11: Used for segmentation of PMCT images. ▪ 3D Slicer version 4.10.2 (BRAINSdemonWarp module): Used for morphing segmented images to the ADAPT FE model. ▪ MATLAB version 2021a: Used for post-processing simulation results, analysing strain data, and validating against autopsy data.
<i>Fiber orientation downsampling compromises the computation of white matter tract-related deformation</i>	Zhou et al.	2022	7	Investigates the influence of downsampling fibre orientation data from diffusion tensor imaging on white matter tract-specific strain. Demonstrates that using a single element-level orientation rather than higher-resolution voxel-wise data can significantly underestimate localised maximum principal strains. Suggests guidelines for improved tract-based modeling fidelity.	▪ LS-DYNA (massively parallel with 128 CPUs): Used for FE simulations. ▪ MATLAB version 2016b: Used for data analysis and post-processing of simulation results.
<i>Brain Material Properties and Integration of Arachnoid Complex for Biofidelic Impact Response for Human Head Finite Element Model</i>	Rycman et al.	2024	0	Presents updated brain tissue properties and an explicit representation of the arachnoid trabeculae in a FE head model to improve biofidelity. Uses recent experimental data on brain tissue to refine a hyper-viscoelastic model and introduces a more realistic brain–skull interface. Though it refines constitutive details, it does not propose an entirely new solver or computational approach.	▪ LS-DYNA version R12: Used for FE simulations (single-precision). ▪ LS-OPT version 6.0.0: Used for optimising hyper-viscoelastic material coefficients. ▪ LS-PrePost version 4.7.9: Used for post-processing simulation data. ▪ CORA version 3.0.0: Used for cross-correlation analyses against experimental data. ▪ Python: Used for custom data processing scripts
<i>A review of brain injury at multiple time scales and its clinicopathological correlation through in silico modeling</i>	Awasthi et al.	2024	0	Surveys in silico modeling techniques across different spatial and temporal scales, highlighting the need for multiscale, multiphysics methods and data integration. Recommends future developments for bridging patient-specific modeling, neurodegeneration, and acute TBI models, but does not itself introduce a new FE method. Instead, it provides a roadmap to integrate existing methods for holistic, personalised TBI simulations.	▪ No specific software: This is a review article discussing multiple existing modeling frameworks rather than implementing a new tool.